• Write the research work and present it in research papers, thesis, conference and journal.	**7. Presentation of the Research (4 hrs)** 7.1. Writing up the research 7.2. Written Presentation 7.2.1. Academic papers 7.2.2. Theses and dissertations 7.2.3. Technical reports 7.3. Visual Presentations 7.3.1. Posters 7.3.2. Infographics 7.3.3. Demonstration 7.4. Oral Presentation 7.4.1. Conference talks 7.4.2. Thesis defense 7.5. Conferences and Journal

1. How does the choice of research presentation format (written, visual, or oral) impact the effectiveness of conveying complex technical-information? Provide examples.

The choice of research presentation format—**written, visual, or oral**—significantly impacts how effectively complex technical information is conveyed, as each format serves distinct purposes and engages audiences in different ways. While written accounts provide the necessary depth for evaluation, visual and oral formats enhance engagement and accessibility for both expert and non-expert audiences.

1. Written Presentations: Documentation and Depth

For most technical researchers, a **written account** is the primary means of evaluation and serves as the definitive documentation of the research process, findings, and conclusions.

- **Impact on Effectiveness:** It allows for the precise, structured, and comprehensive detailing of methods and results that would be too dense for other formats. This is especially critical in **design and creation research**, where a textual report is essential to describe, justify, and contextualize a created technical artifact, such as a new computer-based system.
- **Examples:**
    - **Technical Reports:** These provide a clear and objective account of scientific investigations for specialized stakeholders, often including appendices with raw data and program code.
    - **Journal Papers:** These offer comprehensive, finalized research with in-depth analysis and are considered authoritative sources for the academic community.

2. Visual Presentations: Clarity and Accessibility

Visual presentations are increasingly important because **90% of information transmitted to the brain is visual**, and presenters using visual aids are found to be **43% more effective** in persuading their audience.

- **Impact on Effectiveness:** Visual formats are ideal for simplifying high-dimensional technical data into digestible trends and comparisons. They help narrate the research process through storytelling, making it easier for non-experts to understand technical jargon.
- **Examples:**
    - **Infographics:** These simplify complex research into eye-catching formats that are **30 times more likely to be read** than text-heavy articles, making them highly effective for quick dissemination on websites or social media.
    - **Posters:** Often used at conferences, posters summarize research using minimal text and legible visuals that can be understood from a distance while the researcher discusses the work with delegates.

3. Oral Presentations: Interaction and Persuasion

Oral presentations provide a platform for researchers to explain and justify their work directly to the academic community, allowing for immediate scrutiny and feedback.

- **Impact on Effectiveness:** This format is most effective when it focuses on **highlighting 1–2 main points** or a key argument rather than reading a paper verbatim. The effectiveness of this format depends heavily on the presenter's delivery skills, including speaking slowly, maintaining eye contact, and managing a **Q&A session**.
- **Examples:**
    - **Conference Talks:** Typically 20 minutes long, these are used to present preliminary or ongoing research, allowing for networking and immediate audience feedback.
    - **Software Demonstrations:** These allow researchers to showcase functionalities—such as creating or modifying records—in a step-by-step manner to prove the system works flawlessly.
    - **Thesis Defense (Viva):** A critical oral examination where a student must justify their methodology and mastery of the topic to a panel of examiners.

Comparison Summary

Format	Primary Impact on Effectiveness	Best Use Case
Written	Provides **tangible evidence** and depth; allows for replication by other researchers.	Documenting detailed experiments or comprehensive project work (e.g., dissertations).
Visual	Simplifies data; improves **retention** and accessibility for non-experts.	Presenting trends, comparisons, and key findings at a glance (e.g., posters).

Oral	Facilitates **engagement** and immediate interaction through Q&A.	Presenting preliminary findings or defending a research contribution.

2. Explain the essential qualities of a good research report. Outline and describe the critical steps in developing a research proposal.

A good research report and a well-structured research proposal are both defined by their **logical rigor, clarity, and adherence to established academic standards**.

Essential Qualities of a Good Research Report

- **Focus and Conciseness:** A good scientific paper is **focused on specific questions** and provides adequate evidence for the answers. The writing should be **concise**, as "less is more" when trying to avoid reader confusion.
- **Consistency:** Researchers must **define terms clearly** and use them consistently throughout the report to maintain a professional style. This includes using the same terms for variables in the title, purpose statement, and headings.
- **Coherence:** The report must demonstrate a high level of **coherence**, where every sentence is a logical sequel to the one before it and paragraphs are tied together using techniques like the "hook-and-eye" exercise.
- **Clarity and Objectivity:** Reports should prioritize the **active voice** for vigor, while using the **past tense** to report results and the **present tense** for conclusions. Information should be presented objectively, letting the facts speak for themselves without bias.
- **Standalone Components:** The **abstract, tables, and figures** must be understandable on their own, without requiring the reader to go through the entire paper.
- **Accuracy:** Findings and data must be **accurate and verifiable**, often supported by visual aids like graphs and diagrams to enhance stakeholder understanding.

Critical Steps in Developing a Research Proposal

1. **Identify a Topic and Draft a Title:** The process begins by selecting a subject matter and drafting a **working title** early to ground the study and provide a tangible sign of what is being explored.
2. **Evaluate Researchability:** Reflect on whether the topic **can be researched** (access to participants and resources) and **should be researched** (whether it adds something new to the body of knowledge).
3. **Conduct a Literature Review:** This step involves searching for related studies to find **research gaps**, establish the study's importance, and provide a framework for comparing results later.
4. **Select a Research Design:** The researcher must choose a plan (qualitative, quantitative, or mixed methods) based on the intersection of their **philosophical worldview**, strategies of inquiry, and specific methods.

5. **Anticipate Ethical Issues:** Active planning for **ethical practices**—such as protecting participant anonymity, obtaining informed consent, and disclosing sponsorship—must occur throughout the design process.
6. **Write the Introduction:** A scholarly introduction should use a **narrative hook** to pique interest and employ a **deficiencies model** to justify the study by pointing out shortcomings in past research.
7. **Formulate the Purpose Statement:** This is the most important statement in the proposal, establishing the study's **central intent or controlling idea**.
8. **Develop Research Questions and Hypotheses:** These narrow the purpose statement into **testable signposts** for data collection and analysis.
9. **Detail the Methodology:** The proposal must specify the **methods and procedures**, including the population and sample, data collection instruments, and the specific analysis steps that will be taken.

3. How do you approach academic writing? Discuss the basic structure and potential contents of a research proposal.

Academic writing is a rigorous process that is fundamentally viewed as **"thinking on paper,"** where the act of writing helps researchers clarify their thoughts and discover their true perspectives. It is not merely an end-product but a core research skill that requires consistent practice, feedback, and revision. A research proposal, which serves as the blueprint for this process, is essentially a **critical argument** supported by evidence, aimed at convincing readers that the proposed study will create new knowledge through an appropriate academic process.

Approaching Academic Writing

To approach academic writing effectively, researchers should focus on the following strategies:

- **Establish a Routine:** Writing should be a regular, continuous activity rather than something done in "binges". Successful writers often chart their daily activities to find dedicated blocks of time, writing when they are most fresh and in a place free of distractions.
- **Prioritise Progress Over Perfection:** Early in the process, the focus should be on getting ideas down on paper rather than polishing prose. Experienced researchers often go through **20 to 30 drafts** for a single paper, following a three-stage model: outlining, drafting (shifting and sorting ideas), and finally polishing sentences.
- **Enhance Readability:** Clear academic writing uses **active voice** as much as possible to create vigor. Coherence is maintained through the **"hook-and-eye" technique**, ensuring the key idea of one sentence connects logically to the next. Additionally, researchers must define key terms and use them consistently throughout the manuscript to avoid confusing the reader.
- **Avoid "Fat":** Writers should edit drafts to "trim the fat," which refers to unnecessary words, piled-up modifiers, or excessive prepositions that do not add meaning.

Basic Structure of a Research Proposal

The structure of a proposal varies depending on the research design (qualitative, quantitative, or mixed methods), but it generally follows a conventional pattern:

1. The Beginning (Front Matter)

- **Working Title:** A brief road sign (ideally 12 words or less) that grounds the study and communicates its central notion.
- **Abstract:** A concise summary of the study's purpose, methods, and expected results.

2. The Introduction (The Frame)

The introduction sets the stage and establishes the framework for the research. It often employs a **"deficiencies model"** consisting of five parts:

- **Narrative Hook:** An opening sentence designed to engage the reader and stimulate interest.
- **Research Problem:** A clear statement of the issue or concern leading to the study.
- **Literature Review/Review of Studies:** A brief summary of existing research that justifies the importance of the study and sets it within an ongoing dialogue.
- **Deficiencies in Past Literature:** Identifying gaps, such as topics that have not been explored with a particular population or methodological flaws in previous work.
- **Significance of the Study:** Describing how the research will profit specific audiences and its potential impact on practice or policy.

3. Purpose Statement and Research Questions

- **Purpose Statement:** The most important statement in the proposal, it establishes the study's central intent.
- **Research Questions and Hypotheses:** These narrow the purpose. **Quantitative** studies use questions about variable relationships or hypotheses (predictions), while **qualitative** studies use broad central questions to explore a phenomenon.

4. Methodology (The Plan)

This section details the procedures for conducting the research:

- **Research Design:** Identifying whether the study is experimental, exploratory, or descriptive.
- **Population and Sampling:** Specifying how participants will be selected (e.g., random sampling for quantitative or purposeful sampling for qualitative).
- **Instruments and Materials:** Describing the tools used for data collection, such as surveys, interview protocols, or observation checklists.
- **Data Analysis:** Outlining the steps for analyzing information, whether through statistical procedures for numeric data or thematic coding for text/image data.

5. Ethical Considerations

Proposal writers must anticipate and address ethical dilemmas, including obtaining **informed consent**, ensuring **confidentiality**, and protecting the rights of human participants through Institutional Review Board (IRB) approval.

6. End Matter

- **References:** A list of all sources cited, formatted according to a specific style guide (such as IEEE or APA).
- **Appendices:** Supplementary materials like instruments, interview questions, a timeline for completion, and a proposed budget.

4. As a researcher, how do you prepare and deliver a research presentation? Explain.

Preparing and delivering a research presentation is a critical final step in the research process, focusing on communicating your work effectively to the academic community for evaluation and feedback. Whether for a conference, a thesis defense, or a classroom setting, the process involves careful planning of content, visual aids, and delivery techniques.

Preparing the Presentation

- **Identify the Core Message:** Avoid reading your research paper verbatim; instead, use slides to highlight one or two main points, such as a key argument or a specific aspect of the findings you are particularly proud of.
- **Manage Time and Scope:** Most conference presentations are **20 minutes long followed by a 10-minute Q&A session**. For a 20-minute talk, aim for **10 to 14 slides** to avoid rushing.
- **Design for Readability:** Use a **font size of at least 24 points** to ensure the audience can read your text. Incorporate visuals such as screenshots, software interfaces, or infographics to simplify complex data and enhance engagement.
- **Technical Logistics:** Bring your presentation on a storage medium like a USB and **name your file uniquely** (e.g., your name followed by the conference title) to avoid confusion. Load your file onto the conference system during a break to verify it works as intended.
- **Prepare Handouts:** If your presentation includes complex diagrams or code that may be hard to read on a screen, provide printed handouts for the audience.

Delivering the Presentation

- **Effective Delivery:** Speak **slowly, clearly, and loudly**. Rather than looking constantly at the screen or your computer, maintain **eye contact** with different sections of the audience to build a connection.

- **Use Prompts:** Instead of reading from a script, use a printed copy of your slides with brief notes to stay on track.
- **Handle Nervousness:** Even experienced researchers feel nervous; acknowledging this can often make the audience more sympathetic.
- **Strict Time Management:** Respect the allotted time and watch for signals from the session chair, as overrunning is unfair to subsequent speakers.

Handling the Question and Answer (Q&A) Session

- **Prepare for Scrutiny:** The Q&A is an opportunity to justify your choices and demonstrate your mastered understanding of the topic.
- **Admit Unknowns:** If you cannot answer a question, it is acceptable to say so professionally. You might suggest discussing the point later over coffee or ask the questioner for references on the topic.
- **Guide the Discussion:** You can use your final slide to suggest specific areas where you would welcome audience feedback or further questions.

Alternative Presentation Formats

- **Posters:** These should use eye-catching visuals and **minimal text** to summarize research, with the researcher available at scheduled times to discuss the work with delegates.
- **Software Demonstrations:** Ensure all plug-ins and databases are updated and have the software running before the audience arrives. Always have a **backup plan** to explain what the software should have done if it malfunctions during the demo.
- **Oral Defense (Viva):** This involves a deep-dive discussion with examiners where you must justify your research strategy, methods, and contribution to knowledge.